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Instance-Level Multi-Watermark Embedding with Mask-gated Adapter against Partial Screen-shooting

Mingyue Chen, Xin Liao, Zhen Xia, Chengyu Wang, Xiaoshuai Wu, Yufeng Wu

***Abstract*—Screen-shooting watermarking serves as an effective copyright protection mechanism, enabling watermark extraction from unauthorized captures for leak tracing. Although recent studies have improved robustness to partial screen-shooting or explored local watermark embedding, they do not explicitly support protection of semantic instances. In practice, infringers may capture only valuable regions of an image, such as specific instances of interest. Moreover, different instances within the same image may originate from different sources, making image-level watermarking insufficient for real-world scenarios. To address this, we propose an instance-level watermarking framework that is resilient to partial screen-shooting and enables controllable embedding of multiple watermarks into distinct instances. Specifically, during encoding, we design a mask-gated adapter that controls the regions of watermark embedding. By modulating the number of input mask-watermark pairs, the framework flexibly embeds multiple watermarks while preventing cross-region interference, thereby achieving instance-level protection. During decoding, a locator identifies watermark regions, while a parallel instance segmentation network separates adjacent instances. Finally, the outputs of the two localization branches are fused to obtain precise watermarked instance regions for subsequent watermark extraction, enhancing robustness against partial screen-shooting. Experimental results demonstrate that our framework achieves superior imperceptibility and robustness, particularly in instance-level partial screen-shooting scenarios where the watermarked region covers less than 50% of the image, yielding an improvement in accuracy of more than 10% over that of existing methods.**

***Index Terms*—Robust watermarking, partial screen-shooting, instance-level multi-watermark.**

I. Introduction

With the rapid proliferation of electronic devices such as computers and smartphones, unauthorized capture of screen-displayed content with cameras has become a major threat to digital copyright protection. Digital watermarking provides an effective solution by embedding imperceptible markers (e.g., identification information) into the content for source tracing. Conventional watermarking methods are designed mainly to resist digital distortions such as JPEG compression and Gaussian noise [1]–[4]. In contrast, the screen-shooting process involves cross-media transmission between the digital and optical domains, which introduces far more complex distortions.

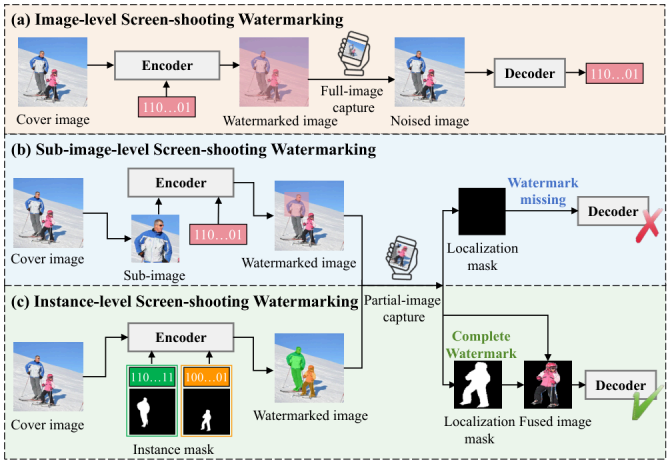


Fig. 1. Illustrations of existing screen-shooting watermarking frameworks and ours. Different colored markers in the watermarked image indicate different watermark regions. Existing methods can be divided into two categories: image-level and sub-image-level watermarking. The former protects the entire image and is suitable for full-image capture, while the latter embeds a watermark in a rectangular sub-image, which may fail to decoded from partial captures due to information loss. In contrast, our method embeds watermarks within irregular instance regions in a mutually isolated manner, enabling instance-level protection and robustness against partial screen-shooting.

Existing screen-shooting watermarking methods are primarily designed for image-level protection and can be broadly categorized into two types: image-level watermarking [5]–[10] and sub-image-level watermarking [11]–[13]. For image-level watermarking, the watermark is embedded and extracted across the entire image, as illustrated in Fig. 1(a). These methods primarily focus on modeling screen-shooting noise to enhance robustness, either by training with real captured datasets or by employing mathematical simulation techniques. In contrast, sub-image-level watermarking embeds watermarks into selected sub-regions of the cover image. To decode such watermarks, a locator network is required to identify the watermarked sub-region from the captured image, as shown in Fig. 1(b). Although these methods have achieved promising results, they rely on full-image capture. When only part of the image is captured, some watermark information is inevitably lost, resulting in decoding failure. Consequently, these methods lack robustness against partial screen-shooting.

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In practice, partial capture is prevalent. Infringers often capture only valuable content regions, such as a single product in an e-commerce photo, a confidential figure in a presentation, or a person of interest in a group image. When only a subset of the scene is recorded, global watermark signals become fragmented, making it impossible to extract or verify. Moreover, modern digital images frequently combine multi-source content, for example, composite AI-generated scenes, advertisement collages, or group photos containing multiple individuals. Different instances within a single image may belong to different contributors. In such scenarios, treating the entire image as a single protection unit is clearly inadequate. These realities highlight the urgent need to shift from image-level watermarking to fine-grained instance-level protection, where each semantically meaningful instance carries its own verifiable watermark, as illustrated in Fig. 1(c). Such capability is essential not only for robustness under partial screen-shooting but also for accountability and traceability in modern multi-source visual ecosystems.

Although recent attempts have explored partial capture, methods such as RoPaSS [14] and FPSMark [15] focus on improving robustness by introducing spatial-layout redundancy across the entire image. RoPaSS introduces symmetric watermarks to increase watermark redundancy. Similarly, FPSMark embeds the same watermark into uniformly distributed sub-images across the whole image, so that robust decoding remains possible as long as at least one complete sub-image is captured. However, these methods do not explicitly consider semantic instances in the image. In contrast, WAM [16] and MaskMark [17] explore local or instance-related watermarking in digital-channel scenarios, but they are still insufficient for our task. Specifically, both methods are designed for digital distortions rather than the screen-shooting scenario. Moreover, when multiple different watermarks need to be embedded into one image, they rely on repeated encoding and region composition. In addition, MaskMark can separate only spatially disjoint watermark regions during decoding. Overall, these methods do not address the task of robust instance-level multi-watermarking under partial screen-shooting.

Achieving reliable instance-level protection against screen-shooting lies in the coupling between spatially selective embedding and robustness to instance-level partial screen-shooting. Specifically, it involves the following non-trivial technical challenges: 1) precise and scalable embedding. Real-world instances often exhibit irregular shapes, and only a subset of instances may require protection. A practical solution must confine watermark signals strictly within irregular instance regions to keep ownership information isolated and traceable, while preventing interference among adjacent instances. Additionally, it should allow flexible embedding of a variable number of watermarks according to user needs, ensuring a controllable and scalable embedding mechanism. 2) Robust instance-level decoding. The cross-media distortions introduced by screen-shooting will significantly disrupt spatial consistency and watermark integrity. A robust watermark decoding process requires joint instance separation and region localization under distortions to achieve effective watermark extraction.

This paper proposes an instance-level watermarking framework designed to resist partial screen-shooting. To achieve precise and scalable embedding, we propose a mask-gated adapter that enables precise and region-aware embedding of distinct watermarks into corresponding instance areas. By gating the embedding process with instance masks, the adapter confines watermark signals within the target regions and enables controllable and scalable multi-watermark embedding. To enable robust instance-level decoding, we develop a parallel extraction architecture consisting of two complementary branches. The first branch acts as a locator, identifying potential watermarked regions from partially captured images. The second branch employs a pretrained instance segmentation network that separates neighboring instances with clear boundaries. By fusing the outputs of the localization and segmentation masks, the framework achieves precise instance-level watermark region recovery, ensuring reliable decoding even under partial capture.

1) We propose a new task of robust instance-level multi-watermarking under partial screen-shooting. To the best of our knowledge, this is the first work to formulate this task and provide an effective framework for it. By embedding watermarks into individual instances, the source of each instance can be verified even when only part of the image is captured.
2) We design a mask-gated adapter that embeds watermarks into specified instance regions. By adjusting the input mask-watermark pairs, it enables controllable embedding of multiple watermarks into different instances in a single forward pass.
3) We introduce a parallel watermark extraction architecture that combines a locator with an instance segmentation network. It enables precise localization and independent decoding of watermarked instances, even when adjacent instances carry different watermarks.
4) Extensive experiments demonstrate that our method outperforms existing approaches in partial capture and multi-watermark scenarios, maintains robustness under diverse shooting conditions, and exhibits strong cross-dataset generalization.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the proposed framework, including the design of the modules and the training strategy. In Section IV, we present several experimental analyses. Section V discusses the applicability boundaries, limitations, and future directions of the proposed framework. Finally, Section VI concludes our work.

II. Related Work

In this section, we review prior work on robust watermarking, focusing on resilience to digital and screen-shooting distortions, and further discuss instance segmentation networks, which serve as a key technique for our instance-level watermarking framework.

A. Digital distortion resilient watermarking

Digital distortion resilient watermarking methods fall into two categories: traditional and deep learning-based methods.

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Traditional methods embed watermarks by directly modifying pixel values (e.g., histogram-based embedding [18]) or applying frequency-domain transformations, such as the Discrete Cosine Transform (DCT) [19], [20], Discrete Fourier Transform (DFT) [21], and Discrete Wavelet Transform (DWT) [22]. However, these approaches rely on hand-crafted features, are robust only to specific distortions and lack generalizability.

Deep learning-based methods have significantly improved robustness to digital distortions. Zhu *et al.* [23] introduced the first end-to-end deep learning framework for robust watermarking, consisting of an encoder, differentiable noise layer, and a decoder. Building upon this, Liu *et al.* [24] proposed TSDL, a two-stage training scheme where the encoder-decoder is first trained without noise and later fine-tuned for various distortions. To improve robustness to JPEG compression, Jia *et al.* [2] introduced MBRS, which randomly mixes real JPEG images and differentiable approximations during training. Ma *et al.* [25] further combined reversible and irreversible embedding mechanisms to enhance robustness. However, these methods are primarily designed for digital distortion and struggle to cope with screen-shooting scenarios.

B. Screen-shooting resilient watermarking

Screen-shooting introduces complex, hard-to-model distortions, making watermark robustness particularly challenging [26]–[28]. Early work by Fang *et al.* [5] applied the scale-invariant feature transform (SIFT) algorithm to identify suitable regions for watermark embedding. Wengrowski *et al.* [7] and Guo *et al.* [10] constructed a real screen-shooting dataset and trained a distortion network to improve real-world applicability. However, collecting real data is time-consuming and device-specific, which limits generalization. To overcome this, many works have simulated screen-shooting distortions. Tancik *et al.* [6] proposed StegaStamp, modeling screen-shooting as a sequence of image operations. Similarly, Fang *et al.* [9] introduced PIMoG, reducing the process to four key distortions: perspective, illumination, moiré, and Gaussian noise. Li *et al.* [29] further designed a grayscale deviation model to handle intensity bias under varying viewpoints. Despite these efforts, most methods operate at the image-level, embedding and decoding watermarks from the entire image. As a result, they fail when only a partial image is captured.

Some sub-image based methods were proposed that introduced a locator into the traditional encoder-decoder architecture. For example, Jia *et al.* [11] and Zhu *et al.* [12] randomly selected a sub-image for embedding, and used a localization network to identify the watermarked region at decoding. He *et al.* [30] embedded distinct watermarks into different image instance regions by repeatedly performing the encoding process. To resist partial captures, Ma *et al.* [14] and Chen *et al.* [15] proposed RoPaSS and FPSMark respectively, which introduced redundant watermarks to enhance robustness. However, these methods only addressed spatial locality while ignoring semantic content, and thus cannot provide instance-level protection.

C. Instance segmentation network

To enable instance-level watermarking, reliable identification of individual instances is required. Instance segmentation provides this capability and has been extensively studied. Top-down methods, such as Mask R-CNN [31], first detect instances and then segment them within bounding boxes, with many follow-ups adopting similar strategies. In contrast, bottom-up methods treat instance segmentation as a label-then-cluster problem. SOLO [32] eliminates the need for bounding boxes, and its improved version SOLOv2 [33] has achieved breakthrough performance. Similarly, CondInst [34] is a one-stage method that generates instance masks using dynamic convolution. In our framework, the locator can identify the regions containing watermarks but cannot distinguish between different instances. Instance segmentation complements this limitation by providing precise instance-level discrimination. Consequently, recent advances in instance segmentation provide a strong foundation for instance-level multi-watermark embedding and extraction, enabling robust protection against partial screen-shooting.

III. Proposed Framework

The proposed framework is illustrated in Fig. 2 and consists of six main components: the encoder E_{nc}, adapter A_{da}, noise layer N, decoder D_{ec}, locator L_{oc}, and an instance segmentation network S_{eg}. Each component plays a distinct role in the overall pipeline. Specifically, E_{nc} and A_{da} are responsible for embedding watermarks into designated instance regions, while the noise layer simulates screen-shooting distortions. D_{ec} is optimized to recover embedded watermarks from distorted images. Simultaneously, L_{oc} localizes potential watermarked regions, and S_{eg} provides instance-level masks to facilitate precise decoding.

For the instance segmentation module S_{eg}, we adopt a pre-trained model rather than joint training. Instance segmentation is a mature field that achieves strong performance on standard benchmarks. Furthermore, since semantic structures remain largely invariant under screen-shooting distortions, a pre-trained model can provide reliable guidance without increasing training complexity.

Jointly optimizing the remaining components is challenging due to their diverse objectives. To address this, we adopt a progressive two-stage training strategy. In Stage I, the encoder E_{nc}, adapter A_{da}, and decoder D_{ec} are jointly trained to achieve robustness against screen-shooting distortions while preserving the high visual fidelity of watermarked images. In Stage II, the pre-trained encoder and adapter are employed to generate multi-watermark images for training the locator L_{oc}, which identifies the watermark regions within the noised image. During the inference stage, a pre-trained instance segmentation network S_{eg} operates in parallel with the L_{oc} to generate instance masks and watermark mask, respectively. These masks are then fused to accurately isolate individual watermarked instances, which are subsequently decoded by D_{ec} for final watermark recovery.

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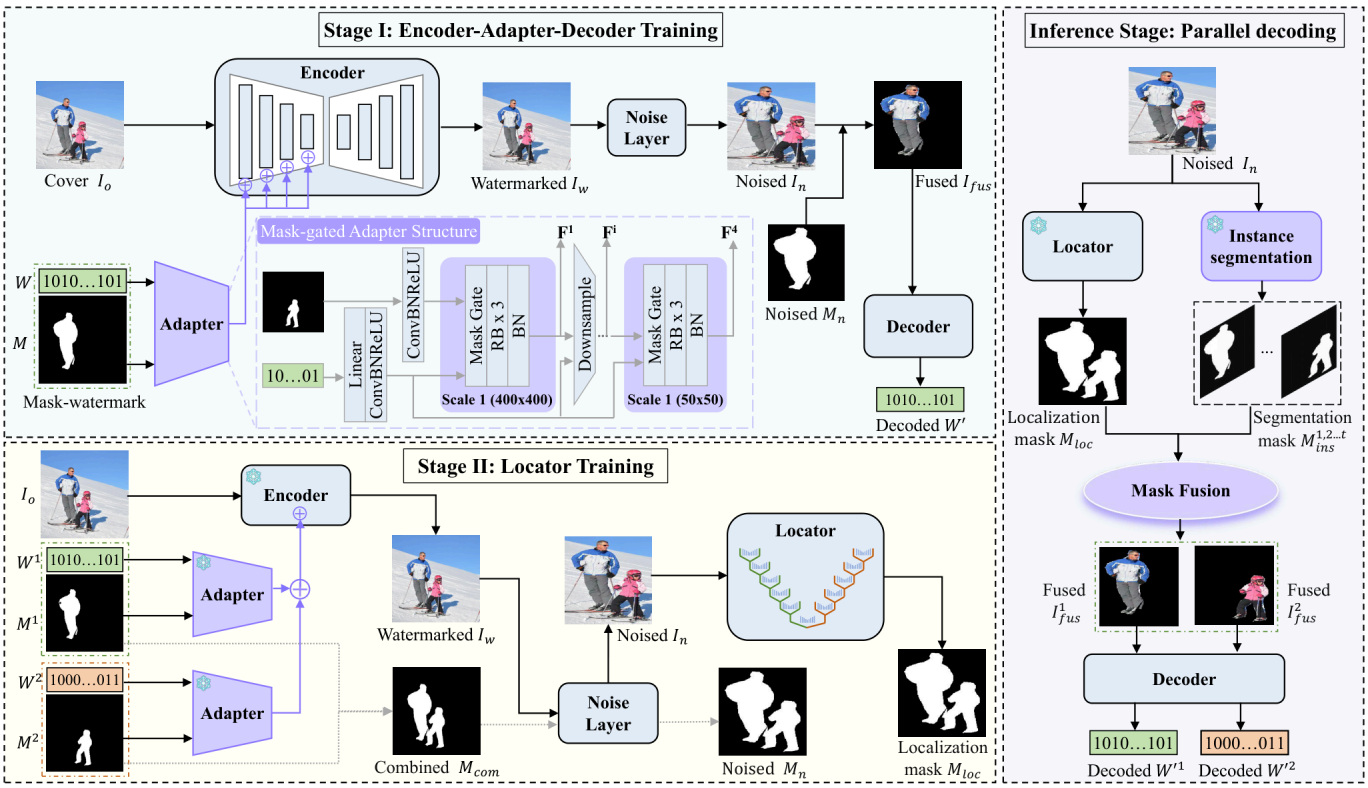


Fig. 2. The framework of the proposed method, which contains two training stages and an inference stage. Stage I jointly trains the encoder, adapter, and decoder for instance-level watermark embedding and robust decoding under screen-shooting distortions. Stage II trains the locator to identify watermark regions using instance-level multi-watermark images. During inference, a pre-trained instance segmentation network runs in parallel with the locator to determine which instances contain watermarks for decoding.

A. Stage I: Encoder-Adapter-Decoder Training

In this stage, the encoder, adapter, and decoder are jointly trained to achieve screen-shooting robustness while preserving image quality. Given a cover image and mask-watermark pairs, the adapter guides the encoder to embed watermarks in the specified regions, producing watermarked images that are subsequently fed into the noise layer and decoder for robust training.

Within this training stage, the adapter plays a crucial role in instance-level embedding. Directly embedding multiple watermarks into different instance regions is challenging for existing encoders. Recent advances in controllable AI image generation, such as ControlNet [35], T2I-Adapter [36], and IP-Adapter [37], provide key inspiration. These methods demonstrate that injecting additional control conditions (e.g., sketches or depth maps) into diffusion models enables precise guidance of the generation process. Inspired by this, we treat watermark embedding in different instance regions as a control condition that guides the encoder to encode watermarks in specific regions. Accordingly, we propose a mask-gated adapter, which directs the encoder to embed distinct watermarks into specific instance regions of cover images $I_o \in \mathbb{R}^{3\times h\times w}$ during image reconstruction, producing instance-level encoded images I_{enc}.

*1) **Mask-gated Adapter**:* The adapter A_{da} is designed to guide watermark embedding in specific regions across multi-scale feature spaces, enabling the isolation of watermarks in different regions. Our mask-gated adapter is illustrated in Stage I of Fig. 2. Given an input mask $M \in \mathbb{R}^{1\times h\times w}$ and watermark W of length l, the watermark is first expanded through a Linear layer and reshaped into a 2D feature map, while the mask is pre-processed by a 3×3 convolution, batch normalization, and ReLU activation (denoted as ConvBNReLU). These features are subsequently passed through four feature extraction blocks with three downsample blocks. At each scale, the watermark features are refined by a Mask Gate (MG) module, which consists of a convolutional layer followed by a Sigmoid activation to generate a spatial gating map. Specifically, the watermark features are modulated by the mask through element-wise multiplication, ensuring that watermark signals are injected at the active regions defined by the mask. Consequently, the MG module guarantees that watermark features are embedded within the designated instance region. Following the MG, three residual blocks and a Batch Normalization layer are applied to enhance the fused features. Finally, four multi-scale mask-watermark fused features $F = \{F^1, F^2, F^3, F^4\}$ are generated, whose dimensions are aligned with the intermediate features of the encoder.

Mathematically, the multi-scale fusion process is defined as:

$$X_i = Conv(X_{i-1}) + G(M_i) \odot Conv(F_{w,i}) \tag{1}$$

where i denotes the i-th scale, X_{i-1} is the feature from the previous scale, $F_{w,i}$ is the watermark feature at scale i, and $\odot$ indicates element-wise multiplication. Here $G(M_i)$ denotes

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a soft mask gate designed to alleviate the artifacts introduced by a hard binary mask:

$$G(M_i) = \sigma(Conv(M_i)) \tag{2}$$

where $\sigma(\cdot)$ is the Sigmoid activation, constraining the gating weights to the range $[0, 1]$. This design smooths the mask boundary and ensures that watermark features are injected predominantly into the target region.

*2) **Encoder**:* Built upon a U-Net backbone [38] with strong reconstruction capability, the encoder injects the mask-watermark fused features F into each downsampling stage by element-wise addition with the intermediate representations, producing the encoded image I_{enc}. However, the differentiable mask-gated adapter generates a soft mask rather than strictly binary one, causing regions intended to be suppressed to exhibit partial activation. This may lead to slight spatial spillover of watermark signals beyond the instance boundaries. To address this issue, we introduce a mask-guided JND (Mask-JND) module to further constrain the watermark residual within instance regions and ensure perceptual invisibility, resulting in the final instance-level watermarked image I_w. Specifically, the proposed Mask-JND extends the enhanced JND model proposed by Wu *et al.* [39], in which the watermark residual is adaptively scaled according to local luminance and contrast characteristics. This mechanism enables stronger embedding in visually insensitive regions while suppressing distortions in perceptually sensitive areas. Moreover, the binary mask is reintroduced during the JND-based refinement process to explicitly constrain the watermark region, thereby enforcing a hard spatial boundary. The process is formulated as follows:

$$I_w = I_o + \mu \times M \times JND\left(I_{enc} - I_o\right) \tag{3}$$

where M is the input mask and μ is a scaling factor controlling the watermark strength, which is set to 1.5 by default. This design guarantees precise instance-level watermarking while preserving image quality.

By combining the MG of mask-gated adapter with Mask-JND perceptual refinement, the proposed framework effectively maintains spatial isolation among multiple watermarks and mitigates cross-instance signal leakage.

*3) **Screen-shooting Distortion Layer**:* To enhance robustness in real-world scenarios, we introduce a screen-shooting distortion layer between the encoder and decoder. Since screen-shooting distortions are complex and difficult to model, we approximate this process through several differentiable noise operations within an end-to-end training framework. Each operation is designed to emulate a typical degradation pattern observed in practical screen-shooting scenarios, as detailed below.

- **Perspective distortion.** When the handheld camera is not parallel to the screen, the image undergoes perspective distortion. Following [6], we simulate this by applying random homography transformations, perturbing the four image corners within 10% of the image size.
- **Crop.** Partial capture frequently occurs during screen-shooting. We randomly crop a region of size $c_1 h \times c_2 w$ with $c_1, c_2 \in [0.5, 1]$ and resize it to $h \times w$ to simulate this degradation.
- **Blur.** Screen-shooting may produce defocus due to camera shake or imperfect focus. Gaussian blur provides a differentiable approximation of such low-pass effects, with kernel sizes randomly sampled from $\{3, 5, 7\}$.
- **Gaussian noise.** To simulate sensor noise and signal fluctuations introduced by the camera and display devices, we add zero-mean Gaussian noise with a variance randomly sampled from $[3, 10]$ to the watermarked image.
- **Color transformation.** Variations across electronic devices, such as differences in resolution and color gamut, can lead to color distortions, including brightness, contrast, and saturation adjustments. We randomly perturb one of these properties, with its magnitude sampled from $(0.5, 1.5)$.
- **Moiré distortion.** Mismatches between the camera sensor and the screen pixel array may produce moiré patterns. We simulate this using the classic algorithm of [9], which generates irregular stripe patterns.
- **JPEG compression.** Imaging and storage further introduce JPEG artifacts stemmed from common compression. We simulate it using the hybrid strategy in [2], combining real and differentiable JPEG compression, with quality factors randomly selected from $[75, 100]$.
- **Illumination distortion.** Environmental conditions such as uneven lighting or reflections can cause regional brightness shifts. To approximate the effect of point and line light sources, we employ point spread and line spread distributions, following the approach introduced by [9].
- **Dropout.** Transmission or storage format may cause some pixel loss. We simulate this by randomly replacing a fraction $[0, 0.3]$ of pixels in the watermarked image with those from the cover image.
- **Resize.** Captured images vary in size. We introduce scaling distortions to unify image dimensions before decoding. We apply a random scaling factor in $(0.5, 2)$ and resize back to the original dimension to ensure consistent decoding input.

By incorporating these differentiable distortions into the training process, the robustness of the proposed framework is effectively improved under diverse screen-shooting conditions.

*4) **Decoder**:* The decoder D_{ec} aims to recover the watermark W' from fused image I_{fus}, which is obtained by combining the noised image I_n and its corresponding mask M_n from the noise layer via element-wise multiplication, i.e., $I_{fus} = I_n \odot M_n$. It is constructed upon a U-Net architecture, where the symmetric encoder-decoder pathway progressively captures and integrates multi-scale features, thereby enhancing the joint modeling of local details and global context. Specifically, the fused image I_{fus} is first processed by ConvBNReLU blocks, four downsampling and upsampling operations to obtain intermediate feature maps of shape (c_f, h, w). These features are subsequently interpolated to a fixed shape of $(1, l, l)$ and transformed by linear layers into a bit sequence, yielding the decoded watermark $W^{'}$ of length l.

This instance-level decoding mechanism ensures that multiple watermarks embedded within distinct regions of a single

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image can be independently and accurately recovered, thus ensuring robustness for partial screen-shooting.

5) ***Loss Function****:* To preserve high visual quality, we constrain the perceptual similarity between I_w and I_o using a mean squared error (MSE) loss. Moreover, to encourage more precise embedding within targeted instance regions, different weights are applied to watermark and non-watermark regions:

$$L_{\text{enc}} = \|\gamma(I_w) - \gamma(I_o)\|^2 \odot \Big(\lambda_1 M + \lambda_2(1-M)\Big) \tag{4}$$

where γ is a differentiable non-parametric mapping function from RGB to the YUV space, and λ_1, λ_2 are balancing coefficients set to 2 and 1 by default.

The decoder is optimized with an MSE loss between the decoded and input watermarks:

$$L_{dec} = \left\| W^{'} - W \right\|^2 \tag{5}$$

The total loss for Stage I is a weighted combination of encoder and decoder losses:

$$L_{StageI} = \lambda_{enc} L_{enc} + \lambda_{dec} L_{dec} \tag{6}$$

where λ_{enc} and λ_{dec} are set as 1 and 20 by default, respectively.

B. Stage II: Locator Training

This stage aims to train the locator network L_{oc} to accurately identify watermark regions. Instance-level multi-watermark images are generated using the pre-trained encoder and adapter and then passed through the noise layer to train the locator for robust localization.

1) ***Multi-watermark Embedding****:* Our framework supports embedding multiple watermarks into different instances without additional training. Specifically, multiple mask-watermark pairs are fed into the adapter A_{da}, as illustrated in Stage II of Fig. 2, with two instances used as an example. The adapter produces fused features for each pair, which are aggregated by weighted summation and then integrated with the cover image features $F_{enc} = \{F^1_{enc}, F^2_{enc}, F^3_{enc}, F^4_{enc}\}$ to generate an instance-level multi-watermark image. Mathematically, this process can be expressed as:

$$F = \sum_{n=1}^{N} \omega^n A_{da}(M^n, W^n) \tag{7}$$

$$\hat{F}^i_{enc} = F^i_{enc} + F^i, i \in \{1,2,3,4\} \tag{8}$$

where N denotes the number of mask-watermark pairs, and ω^n controls the contribution of each adapter output. In our method, all instances are treated equally, and we set $\omega^n = 1$. The input masks are required to be spatially non-overlapping to prevent interference among watermark signals across different instances, ensuring that watermark embeddings remain spatially isolated.

2) ***Locator****:* The localization network L_{oc} aims to identify watermark regions from the noised image I_n, which has undergone screen-shooting distortions, producing a corresponding binary mask M_{loc}. Existing sub-image-level watermarking methods have made remarkable progress in localization. For example, Jia *et al.* [11] and Zhu *et al.* [12] employed HRNet to detect the four corner points of a rectangular watermarked sub-image. However, such methods are limited to regular, rectangular sub-images and are unsuitable for irregular instance-level watermark regions in our task. In addition, He *et al.* [30] used an instance segmentation network for localization, assuming that each detected instance might contain a watermark. In contrast, our method embeds watermarks only in specific instances that need protection, which thus requires discriminating which instances are watermarked.

Inspired by FPSMark [15], we address this by leveraging intrinsic image feature differences to distinguish watermarked from non-watermarked regions. Specifically, we adopt U^2-Net† [40] as the locator. It is a lightweight variant of U^2-Net and employs a nested U-structure that preserves the multi-scale feature extraction advantage of UNet, while enhancing feature representation through increased depth and width. This architecture strengthens comprehension of both local and global context and aligns well with our multi-scale watermark encoding process. Its deep encoder effectively captures subtle features introduced by watermarks, while the symmetric decoder recovers fine spatial details. These properties make it particularly suitable for pixel-level localization of irregular watermark regions.

3) ***Loss Function****:* In Stage II, the pre-trained encoder and adapter are frozen. The clean ground-truth combined instance mask $M_{\text{com}} = M^1 \cup M^2 \cup \cdots \cup M^N$ is passed through the same noise layer applied to I_w, producing the corresponding noised mask M_n for supervision.

To ensure precise localization, we adopt a hybrid loss combining binary cross-entropy (BCE) [41] and intersection-over-union (IoU) [42] losses at both pixel and region levels. Furthermore, to mitigate overfitting, we follow the deeply supervised learning strategy introduced in HED [43], applying supervision by computing the loss at multiple decoding layers of the U^2-Net† locator. The overall locator loss, that is, the loss at this stage is defined as:

$$L_{\text{Stage II}} = L_{\text{loc}} = \sum_{k=1}^{K} \alpha_k \left(L^{(k)}_{\text{bce}} + L^{(k)}_{\text{iou}}\right) \tag{9}$$

where $L^{(k)}_{\text{bce}}$ and $L^{(k)}_{\text{iou}}$ are the BCE and IoU loss of the k-th side output, respectively. K denotes the total number of the decoding layers of the U^2-Net†, and α_k is the corresponding layer weight. In our implementation, K is set to 7, including one final layer and six intermediate layers. The weight of the final output is empirically set to 1, while each intermediate output is assigned a weight of 0.25.

C. Inference Stage

In contrast to image-level watermarking methods that extract watermarks from the entire image and sub-image-level methods that localize and then decode from sub-regions, our

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Fig. 3. Visual comparison of watermarked images embedded by different methods. From top to bottom, the number of watermarked instances are 1, 3, and 5. Each row shows the watermarked images and their corresponding residual images (residuals are magnified 5× for better visualization). Our method achieves superior performance, without noticeable artifacts or color distortions.

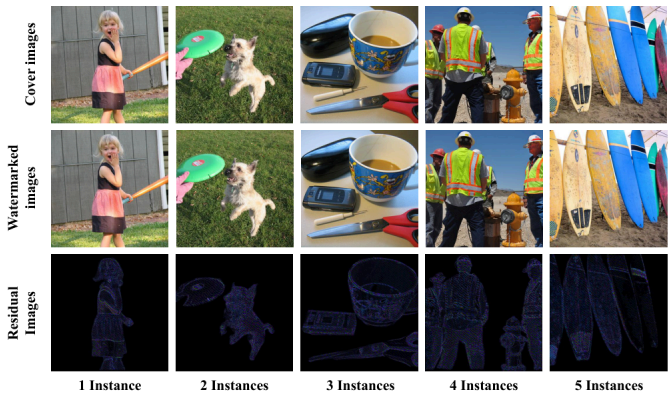


Fig. 4. Watermarked images and residuals, with different numbers of watermarked instances, without mutual interference.

instance-level multi-watermark framework first detects the watermarked instances and then performs decoding. However, the watermark localization network only performs binary classification between watermarked and non-watermarked regions, and cannot distinguish adjacent instances carrying different watermarks. To address this, we introduce an instance segmentation network and a dual-branch mask fusion architecture in the inference stage. This architecture performs two tasks in parallel: a watermark localization branch that detects regions containing watermarks, and an instance segmentation branch that identifies instances and generates their masks. The outputs of the two branches are then fused to produce precise instance-level watermark masks, which are finally fed into the decoder for watermark extraction and copyright verification.

1) ***Instance segmentation network****:* During screen-shooting, distortions such as illumination changes, JPEG compression, and geometric transformations may be introduced. However, the semantic categories and boundaries of instances in the image generally remain unchanged. Leveraging this invariance of high-level semantic information can enhance the robustness of instance identification under such distortions. Therefore, we introduce an instance segmentation network to precisely identify instances and generate a binary mask for each instance.

For this purpose, we employ pre-trained SOLOv2 [33] as the instance segmentation branch S_{eg}. SOLOv2 is a high-performance CNN-based model whose simplicity makes it well-suited for integration with the other modules in our framework. It divides an image into an $S \times S$ grid, where each cell predicts the category and mask of the instance centered within it. Its core design utilizes dynamic convolution for efficient mask generation and matrix non-maximum suppression for fast post-processing. The end-to-end mask prediction of SOLOv2 simplifies data interfacing and facilitates seamless integration with our localization branch. Moreover, its high accuracy and efficiency on standard benchmarks ensure its feasibility and reliability in our watermark extraction task.

This branch takes the noised image I_n as input and outputs a set of instance masks $M_{ins} = \{M_{ins}^i \mid i = 1, 2, ..., t\}$, where t denotes the total number of detected instances. These masks are then fused with the watermark localization mask M_{loc} to determine the final instance-level watermark extraction regions.

2) ***Mask Fusion****:* The locator L_{oc} cannot distinguish individual instances, while the instance segmentation network S_{eg} does not determine which instances contain watermarks. This limitation becomes pronounced when multiple adjacent instances are watermarked, as they may merge into a single connected region.

To overcome these limitations, we introduce a mask fusion module that combines the strengths of both branches: the fine-grained instance masks provided by S_{eg} and the watermark localization mask from L_{oc}. The goal is to identify only those instances that contain watermarks, yielding the final fused masks M_{fus}.

Formally, for each instance segmentation mask M_{ins}^i, we compute its intersection with the localization mask and define

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TABLE I
VISUAL QUALITY OF WATERMARKED IMAGES OF EACH METHOD, WITH DIFFERENT NUMBERS OF WATERMARKED INSTANCES AND DIFFERENT WATERMARK COVERAGE RATIOS

Method	Different numbers of watermarked instances						Different ratios of watermark coverage							
	1 Instance		3 Instances		5 Instances		cover 10%-20%		cover 30%-40%		cover 50%-60%		cover 70%-80%	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
StegaStamp [6]	36.66	0.961	31.15	0.915	29.83	0.893	34.25	0.967	31.02	0.924	29.71	0.895	29.16	0.868
RIHOOP [44]	38.61	0.970	33.41	0.937	31.91	0.918	33.41	0.937	33.25	0.940	32.02	0.920	31.74	0.904
PIMoG [9]	38.95	0.980	33.42	0.940	32.28	0.919	37.55	0.973	34.96	0.960	32.12	0.918	31.89	0.916
FPSMark [15]	**41.82**	**0.987**	**36.68**	0.968	35.91	0.964	39.40	0.982	37.11	0.970	35.64	0.963	35.01	0.954
Ours	40.91	0.985	36.52	**0.969**	**36.00**	**0.964**	**40.24**	**0.983**	**37.22**	**0.971**	**35.73**	**0.963**	**35.09**	**0.955**

the instance coverage:

$$s_i = \frac{|M_{ins}^i \cap M_{loc}|}{|M_{ins}^i|} \tag{10}$$

The score s_i quantifies the overlap between the instance and the watermark region. The final fused masks are then obtained by thresholding s_i:

$$M_{fus} = \{M_{ins}^i \cap M_{loc} \mid s_i > \theta, i = 1, 2, ..., t\} \tag{11}$$

This simple yet effective fusion strategy is invariant to the number and size of watermarked instances. After mask fusion, we obtain a set of precise instance-level watermark masks $M_{fus} = \left\{M_{fus}^i \mid i = 1, 2, ..., n\right\}$. For each mask, the corresponding instance region is isolated from the noised image I_n to form the fused instance image I_{fus}^i, defined as $I_{fus}^i = I_n \odot M_{fus}^i$, where $\odot$ denotes element-wise multiplication. Each fused image is then fed into the decoder for independent decoding, enabling the extraction of the watermark embedded in each instance for copyright authentication.

This parallel decoding mechanism ensures that, even in complex scenes with multiple adjacent instances, the framework can rely on the fused masks to independently extract the watermark of each instance, thereby achieving robust copyright verification under partial screen-shooting conditions.

IV. EXPERIMENTAL RESULTS

This section first introduces the implementation details, followed by comparisons with state-of-the-art methods in terms of visual quality and robustness. This evaluation covers instance-level watermarking with varying numbers of instances and different watermark coverage ratios, under both full and partial screen-shooting scenarios. In addition, we further assess robustness under diverse capture conditions, including shooting distances, angles, and devices, and evaluate the generalizability on other datasets. Finally, ablation studies and comparisons across different segmentation networks are conducted to validate the effectiveness of our framework.

A. Implementation Details

In the training phase, we randomly select 10000 images with corresponding object masks from the COCO dataset [45]. All training images are center-cropped and resized to 400×400. Instance masks are obtained from the COCO annotations and filtered to ensure spatial non-overlap. Specifically, masks with extremely small areas or multiple disconnected components are discarded, and overlapping pixels between masks are removed sequentially to obtain a set of spatially non-overlapping instance masks. The watermark information for each instance is represented as a randomly generated binary sequence of length $L = 100$ bits. Our method is implemented in PyTorch and executed on an NVIDIA RTX 4090 GPU. In the first training stage, the encoder, mask-gated adapter, and decoder are jointly trained for 500 epochs with a batch size of 8, using the AdamW optimizer [46] with a learning rate of 1e-4. In Stage II, the encoder and adapter trained in Stage I are frozen and used to generate multi-instance, multi-watermark images. Each image contains at most five watermarks. The locator is trained for 500 epochs with a batch size of 16 using AdamW, with an initial learning rate of 1e-4 and a cosine annealing schedule [47] decaying to 1e-6. The pre-trained instance segmentation network is used during inference to locate watermarked instances, in parallel with the locator. In the mask fusion step, the overlap threshold θ is set to 0.4.

TABLE II
EXTRACTION ACCURACY UNDER FULL SCREEN-SHOOTING WITH VARYING NUMBERS OF WATERMARKED INSTANCES

Method	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances
StegaStamp [6]	65.51%	82.24%	85.53%	87.94%	92.34%
RIHOOP [44]	68.66%	87.05%	90.18%	91.70%	94.33%
PIMoG [9]	70.83%	85.93%	87.83%	89.4%	91.57%
FPSMark [15]	67.63%	90.21%	92.91%	95.53%	97.41%
FPSMark† [15]	69.82%	91.62%	94.62%	96.56%	98.41%
Ours	**99.67%**	**99.46%**	**99.09%**	**99.05%**	**98.73%**

During the inference stage, we use the COCO validation set. Since the number and size of instances in the images vary, we divide the experiments into two categories: varying numbers of instances and varying watermark coverage ratios. For each experimental group, defined by a specific setting of these factors, 100 images are randomly selected for evaluation. Visual quality is assessed using Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM), while robustness is measured by extraction accuracy (ACC). We compare our method with four representative methods: StegaStamp [6], RIHOOP [44], PIMoG [9], and FPSMark [15]. As these methods are designed for image-level watermarking, we adapt them to instance-level evaluation by multiplying their outputs with binary masks, thereby ensuring fairness in comparison. The default capturing device is iPhone

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Fig. 5. Visual results of the localization, segmentation, and fused masks with different numbers of watermarked instances.

12, and the display is Lenovo XiaoXin Pro 14. During the capturing process, the phone is mounted on a stand and operated via Bluetooth remote control to avoid any deviations caused by touching. All watermarked images are captured under the same conditions in the comparison experiments.

B. Visual Quality

In this section, we compare the visual quality of watermarked images under different numbers of instances and ratios of watermark coverage with other methods. Representative results are reported in Table I, showing that our method achieves consistently high visual quality across all settings. For example, when five instances are watermarked, the PSNR reaches 36.00 dB and the SSIM is 0.964, which are comparable to those of FPSMark [15]. FPSMark embeds watermarks into uniformly distributed image blocks, whereas the images obtained using instance masks do not necessarily contain watermarks throughout the entire mask area. As a result, its visual quality may appear higher, and this difference becomes more evident when only a single instance is present.

Moreover, Fig. 3 presents visual examples of watermarked images and residual images. As observed, StegaStamp [6] introduces noticeable artifacts in smooth regions, while PIMoG [9] exhibits slight color distortions. In contrast, our method better preserves the original visual quality.

In addition, to demonstrate the effectiveness of our method in instance-level watermark embedding, we present visual examples with varying numbers of watermarked instances in Fig. 4, together with their corresponding residuals. Each instance is embedded with an independent watermark. As observed from the residual images, the watermark signals are strictly confined within the designated instance regions without spreading to others, indicating that different instance-level watermarks do not interfere with each other. This property provides a strong guarantee for reliable watermark extraction. Moreover, the robustness experiments in the following section further confirm the isolation of instance-level watermarks.

TABLE III
EXTRACTION ACCURACY UNDER PARTIAL SCREEN-SHOOTING WITH VARYING NUMBERS OF WATERMARKED INSTANCES

Method	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances
StegaStamp [6]	54.95%	60.33%	65.26%	68.03%	81.08%
RIHOOP [44]	52.20%	53.76%	58.05%	59.83%	68.23%
PIMoG [9]	58.37%	59.35%	66.15%	67.68%	75.56%
FPSMark [15]	62.78%	77.75%	83.73%	80.52%	96.33%
FPSMark† [15]	62.98%	81.47%	85.62%	83.26%	97.50%
Ours	**98.40%**	**98.68%**	**98.24%**	**96.15%**	**97.86%**

C. Instance-level Watermarking with Varying Numbers of Watermarked Instances

To evaluate the robustness of our method in the instance-level watermarking setting, we conducted experiments with varying numbers of watermarked instances. In this experiment, each instance is embedded with an independent watermark, and all instances are non-overlapping. The number of watermarked instances ranges from one to five and evaluations are performed under both full and partial shooting scenarios.

To ensure a fair comparison with image-level watermarking baselines, a mask adaptation process is applied to them. Specifically, the full-image watermarked outputs of the baselines are multiplied pixel-wise with the combined binary masks of all target instances, while non-mask regions are replaced with the original image. This adjustment confines watermark embedding to instance regions while preserving the original embedding mechanism and testing pipeline of each baseline. In addition, FPSMark [15] is designed for partial screen-shooting and is the closest baseline to our task. We further fine-tune its decoder and localization network with mask supervision to adapt it to the instance-level decoding setting, and denote the resulting variant as FPSMark†.

1) Full screen-shooting: In this experiment, the watermarked image is enlarged to fill the entire screen, and the smartphone is positioned at a fixed distance of 30 cm from the display to ensure full capture. The captured images are manually corrected for geometric distortions following the same procedure as PIMoG [9], with irrelevant background regions cropped out. They are then resized to $h \times w$ before being fed into the locator and segmentation network. The watermark is extracted from each instance, and the final accuracy is computed as the mean decoding accuracy across all watermarks. The results of different methods are summarized in Table II.

As shown in Table II, our method achieves the best performance, with accuracy exceeding 98% in all cases. The accuracy of baseline methods increases with the number of watermarked instances, since a larger watermark coverage makes the task more similar to full-image watermarking. Nevertheless, our method outperforms all baselines, demonstrating its effectiveness in instance-level multi-watermark embedding.

Fig. 5 presents qualitative examples with different numbers of watermarked instances. The first row shows the captured images with irrelevant borders removed, the second row displays the localization masks predicted by our localization network, the third row shows instance segmentation outputs where

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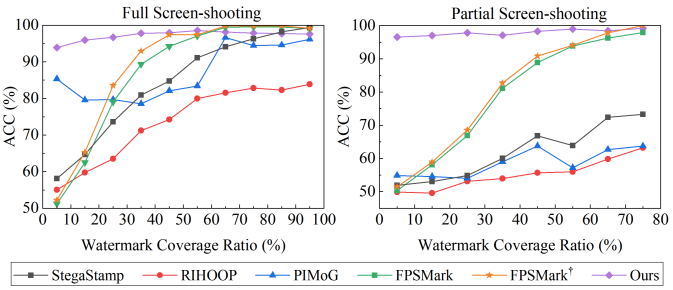


Fig. 6. Watermark extraction accuracy under full and partial screen-shooting with different watermark coverage ratios.

different instances are marked in distinct colors, and the fourth row shows fused masks obtained by combining localization and segmentation results. As shown, when multiple instances are close to each other, the localization network alone struggles to delineate precise boundaries. However, the segmentation network effectively compensates for this limitation, ensuring reliable watermark extraction.

2) Partial Screen-shooting: Unlike the partial shooting experiments in image-level watermarking, which are based on different capture ratios, instance-level watermarking is inherently more complex. Since the number, size, and spatial distribution of instances vary across images, the partial shooting process cannot be easily approximated through proportional cropping. In this experiment, we adjust the position and angle of the smartphone to capture regions containing watermarked instances, with a shooting distance within 30 cm and a capture angle not exceeding 40°. To ensure fairness, all competing methods are tested under the same capture conditions.

As shown in Table III, our method exhibits a clear advantage. Despite the increased difficulty of partial shooting, its performance remains highly consistent with the full screen-shooting results, with only minor reductions in accuracy. When only a single instance is watermarked, the accuracy of the baseline methods drops to around 55%. While FPSMark† shows improvement over FPSMark, it is still less robust than our method.

Overall, our method maintains stable and high decoding accuracy across different numbers of watermarked instances in both full and partial shooting scenarios, providing strong support for practical instance-level copyright protection.

D. Instance-level Watermarking with Varying Watermark Coverage Ratios

Since the sizes of watermarked regions vary across images, we evaluate the robustness under varying watermark coverage ratios. Accordingly, the test set is divided into subsets at 10% intervals according to the watermark coverage ratio. Similar to Sec. IV-C, evaluations are performed under both full and partial shooting scenarios.

1) Full screen-shooting: In the full screen-shooting scenario, the complete watermarked image is captured, resized, and then processed for watermark localization and instance segmentation. The extraction accuracies of the different methods are presented in Fig. 6.

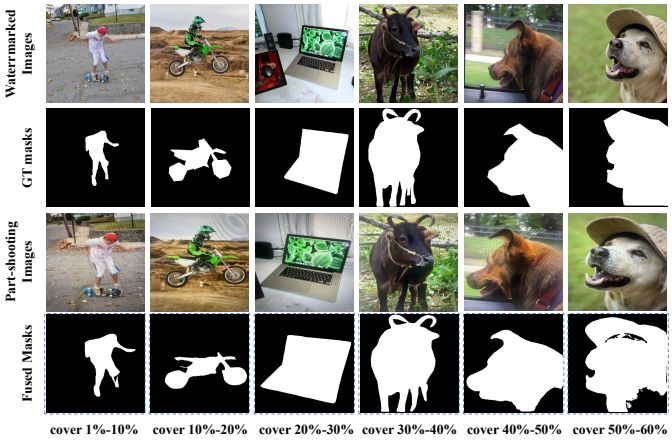


Fig. 7. Visual examples of single-instance watermarked images under partial screen-shooting with varying watermark coverage ratios.

The results show that our method consistently achieves superior performance across all watermark coverage ratios. With the exception of when the watermark coverage is less than 10%, the extraction accuracy remains above 95%, indicating strong robustness. When the watermark region is small, fewer pixels carry the embedded watermark. Therefore, even slight screen-shooting distortions may noticeably reduce the effective recoverable watermark information. In addition, slight localization or segmentation deviations occupy a larger proportion of a small instance, which may further affect mask fusion and decoding accuracy. In contrast, when the watermark coverage is below 30%, the extraction accuracy of baseline methods typically drops below 80%, highlighting their limited robustness in the instance-level watermarking setting. Although FPSMark† is adapted to irregular instances, its robustness still falls short of our method. As the watermark coverage increases, their performance gradually improves. When the coverage exceeds 60%, FPSMark achieves an accuracy comparable to ours, which is attributed to its block-wise redundant encoding strategy. FPSMark is designed to resist partial screen-shooting by embedding redundant watermark information into uniformly distributed image blocks. With higher watermark coverage, the probability that the watermark region overlaps with the predefined image blocks increases, thereby increasing the likelihood of capturing at least one complete watermark block. As reported in the FPSMark paper, when the captured area exceeds 55% of the image, at least one complete watermark block is guaranteed to be preserved, which ensures reliable watermark decoding. At higher coverage ratios, embedding multiple watermarks increases the likelihood of adjacent watermark regions, posing additional challenges for our mask fusion strategy. Nevertheless, our method maintains competitive performance even under such conditions, confirming its robustness and adaptability to complex instance-level scenarios.

2) Partial Screen-shooting: The partial screen-shooting setup follows the same procedure as in Sec. IV-C2. Compared with full screen-shooting, this scenario involves image scaling and loss of peripheral semantic context, making both localization and instance segmentation more challenging. Notably,

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TABLE IV
EXTRACTION ACCURACY WITH DIFFERENT DISTANCES AND ANGLES

Distance	Left 40°	Left 30°	Left 20°	Left 10°	0°	Right 10°	Right 20°	Right 30°	Right 40°
20 cm	96.60%	97.58%	98.40%	98.69%	99.36%	98.42%	98.30%	97.06%	96.82%
30 cm	98.73%	98.80%	99.12%	99.00%	99.37%	98.67%	99.04%	98.65%	97.74%
40 cm	98.62%	98.70%	98.77%	99.35%	99.21%	99.28%	98.55%	98.88%	98.29%

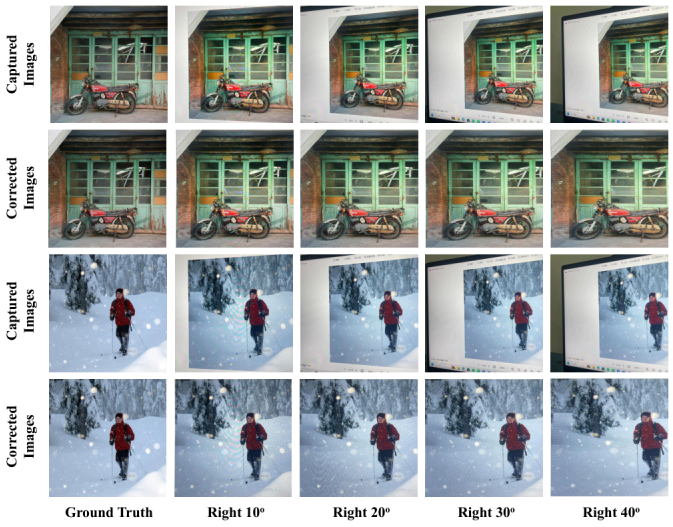


Fig. 8. Representative examples of captured images at a distance of 20 cm under different shooting angles (10°–40°) and their corresponding perspective-corrected images.

TABLE V
EXTRACTION ACCURACY WITH DIFFERENT DEVICES

Device	iPhone 12	Vivo S17t	SAMSUNG S20FE
Lenovo XiaoXin Pro 14	99.67%	97.54%	98.96%
ENVISION G249G	98.67%	97.29%	99.14%
AOC 27E12HM	98.53%	96.72%	98.81%

when the watermark coverage ratio is excessively high (e.g., 80%), partial shooting becomes nearly equivalent to full-screen shooting. Therefore, this experiment is conducted with watermark coverage below 80%.

As illustrated in Fig. 6, our method consistently outperforms all baselines under various partial shooting conditions, achieving an extraction accuracy exceeding 95% across all tested cases. In contrast, the competing methods show limited robustness to partial captures. Specifically, StegaStamp, RIHOOP, and PIMoG fail to decode when the watermark coverage is small, and their accuracy remains below 75% even at relatively high coverage ratios (70%-80%). For FPSMark and FPSMark†, their block-wise redundant embedding strategy provides certain robustness when the coverage ratio is moderate (50%-60%), reaching accuracy above 90%. The watermark region may still partially coincide with one of the predefined image blocks even when the watermark coverage is relatively small (e.g., 35%). However, due to the irregular shapes of instances, the captured region may not align well with the predefined block partition, making it difficult to guarantee the capture of a complete watermark block even at high coverage ratios, which leads to a noticeable decrease in performance. In contrast, our method maintains stable performance across different coverage ratios, achieving reliable copyright protection.

Fig. 7 presents several visual examples. For instance, in the single-instance watermarking case, the first two rows display the watermarked images and their corresponding binary instance masks, while the last two rows show the partially captured images and the fused masks. Under different watermark coverage ratios, our method accurately identifies the watermarked regions from partially captured images. Despite strong moiré artifacts and complex instance shapes (e.g., human limbs, motorcycles), the model can still effectively detect and decode the embedded watermarks. Even in cases where the watermark region is not fully identified, the watermark can still be successfully extracted, benefiting from the dropout noise layer introduced during training, which enhances the model's robustness.

E. Robustness to Different Shooting Conditions

In real-world scenarios, shooting conditions are often complex and diverse. To evaluate the robustness of our method under varying capture environments, we conducted tests across different shooting distances, viewing angles, display devices, and capturing devices. The evaluation is performed on images containing one watermarked instance to comprehensively assess robustness.

1) Different Distances and Angles: Capturing images from different distances and angles is common in practice, often leading to perspective distortion. To evaluate the robustness of our method in such complex scenarios, we conduct experiments at distances of 20 cm, 30 cm, and 40 cm, and at angles ranging from left 40° to right 40° with 10° intervals. These settings represent common shooting conditions in real-world scenarios. Fig. 8 presents representative examples captured at a distance of 20 cm under different shooting angles ranging from 10° to 40°, together with their corresponding perspective-corrected images. Since the size and spatial layout of instances vary across images, we select 100 images containing a watermarked instance to ensure that the target instance could be captured under all the above conditions.

As shown in Table IV, our method maintains high extraction accuracy across the tested shooting distances and angles, achieving over 96.60% accuracy in the evaluated settings. Although larger shooting angles introduce more severe perspective distortion, the method remains robust under the tested practical shooting conditions.

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TABLE VI
VISUAL QUALITY AND EXTRACTION ACCURACY ON THE LVIS DATASET

Metric	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances
PSNR	39.51	37.33	36.71	35.95	36.02
SSIM	0.9811	0.9686	0.9658	0.9593	0.9610
ACC	96.30%	92.17%	98.13%	97.36%	99.34%

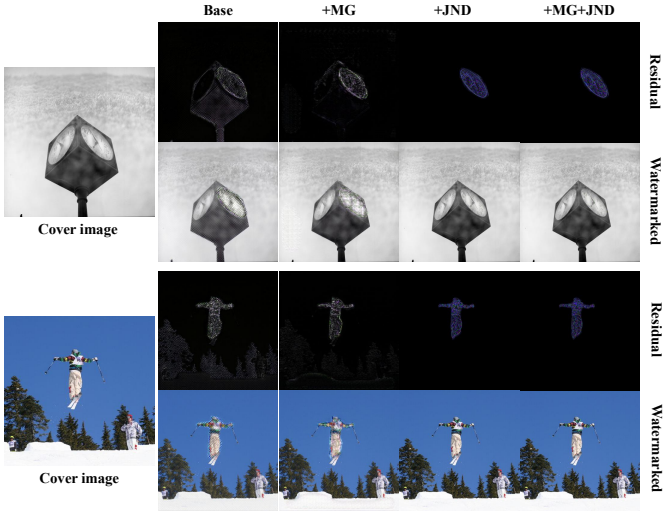


Fig. 9. Visualization comparisons of different instance-specific embedding methods. "Base" denotes the model without the Make Gate (MG) module and JND. The residual images are amplified by a factor of five for better visibility.

2) Different Devices: Different display and capture devices have distinct imaging characteristics, illumination responses, and sensor noise profiles, which can introduce various distortions during the screen-shooting process. To evaluate robustness under device variability, we conduct experiments with three display devices (Lenovo Xiaoxin Pro 14, ENVISION G249G, AOC 27E12HM) and three capture devices (iPhone 12, Vivo S17t, SAMSUNG S20FE). All experiments are performed at a fixed distance of 30 cm with frontal capture. As shown in Table V, our method consistently achieves over 96% extraction accuracy across all device combinations, demonstrating strong generalization and stability in practical cross-device scenarios.

F. Robustness to Cross-Dataset

To evaluate the generalization ability of our method, we conduct cross-dataset experiments on the LVIS dataset [48], which like COCO, contains multiple object instances. Experiments are performed on subsets of images containing different numbers of watermarked instances, with each subset consisting of 50 images. Unlike COCO, the LVIS test set does not provide instance masks. Therefore, we generate masks using SOLOv2 for watermark embedding. This approach also reflects practical scenarios, where instance masks may be obtained after image acquisition.

Table VI presents the visual quality and watermark extraction accuracy of our method on the LVIS dataset. When embedding five instances, our method achieves an extraction accuracy of 99.34% while maintaining high visual fidelity, demonstrating strong cross-dataset generalization.

TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT EMBEDDING AND DECODING METHODS UNDER PARTIAL SCREEN-SHOOTING. "S" AND "P" INDICATE SERIAL AND PARALLEL, RESPECTIVELY.

Method	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances
Base+JND	63.75%	58.70%	60.49%	63.72%	68.13%
Base+MG+JND (S)	87.72%	89.79%	90.00%	89.96%	92.18%
Base+MG+JND (P)	**98.40%**	**98.68%**	**98.24%**	**96.15%**	**97.86%**

G. Ablation Study

This section presents ablation studies to verify the effectiveness of the key components in our framework. We focus on three critical factors: the instance-specific embedding method used during watermark embedding, the parallel extraction architecture employed during watermark extraction, and the role of the instance segmentation branch in decoding.

1) Effect of instance-specific embedding methods: During watermark embedding, the accurate integration of watermark information into instance-specific regions primarily benefits from the MG module and the JND post-processing. To assess their contributions, we conducted ablation experiments by selectively removing either the MG module or the JND operation.

As shown in Fig. 9, the "Base" (without MG and JND) produces evident residual artifacts outside the target instance, indicating that watermark signals are not spatially constrained and tend to accumulate in high-frequency areas. Incorporating the MG module reduces artifacts but still leaves slight traces beyond the instance boundary. After introducing JND, these residuals almost disappear, as the JND operation suppresses watermark energy in non-target regions using the guidance of the instance mask. Notably, both "Base+JND" and "Base+MG+JND" achieved visually ideal results. Therefore, we verified their robustness by adjusting the watermark strength to maintain visual equivalence for a fair comparison. As shown in Table VII, validations are conducted on different numbers of watermarks in partial screen-shooting scenarios. The "Base+JND" shows noticeably weaker robustness, with decoding accuracy significantly lower than that of "Base+MG+JND (P)", where P denotes the parallel architecture. These results demonstrate that the MG is essential for spatially constraining the watermark within target instances, while JND further enhances imperceptibility and suppresses non-target embedding, together achieving the best balance between fidelity and robustness.

2) Effect of Watermark Extraction Architecture: In the watermark extraction stage, our framework employs a parallel architecture that integrates a localization network and an instance segmentation network. This design ensures accurate watermark localization and mask generation simultaneously. To assess the impact of this design, we replaced the parallel structure with a serial one, where localization is performed first, followed by segmentation on the localized image.

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TABLE VIII
EXTRACTION ACCURACY WITH AND WITHOUT THE INSTANCE SEGMENTATION BRANCH

Setting	Full Screen-shooting					Partial Screen-shooting				
	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances	1 Instances	2 Instances	3 Instances	4 Instances	5 Instances
w/ Segmentation	99.59%	99.62%	99.48%	98.99%	98.89%	98.57%	98.86%	98.22%	97.03%	97.60%
w/o Segmentation	97.60%	99.97%	100.00%	100.00%	99.74%	96.04%	99.86%	97.96%	99.91%	100.00%



Fig. 10. Failure cases of instance segmentation with the serial decoding architecture under partial screen-shooting.

We conducted partial screen-shooting experiments on images containing different numbers of watermarks. As shown in Table VII, the serial structure leads to lower decoding accuracy than the parallel version across all cases. Representative failure cases are illustrated in Fig. 10, where the segmentation network incorrectly produces instance masks in regions without meaningful semantic content. This degradation occurs because cropped watermarked images lack surrounding contextual information, which is essential for precise segmentation. Moreover, existing instance segmentation networks are typically trained on full natural images, leading to poor generalization when applied to preprocessed or cropped images. Therefore, the proposed parallel architecture achieves a better balance between semantic integrity and structural consistency, resulting in superior performance in watermark extraction.

3) Ablation on the Instance Segmentation Branch: To further analyze the role of the instance segmentation branch in decoding, we conduct an ablation study under a simplified setting where all watermarked instances share the same watermark. Experiments are conducted with and without the instance segmentation branch under both full and partial screen-shooting scenarios, with varying numbers of watermarked instances.

As shown in Table VIII, when all instances share the same watermark, removing the instance segmentation branch does not lead to a significant performance drop. Under both full and partial screen-shooting, the decoding accuracy without the segmentation branch remains comparable to that with the branch, and is even slightly higher in several multi-instance settings. In the shared-watermark setting, the localization branch alone can already provide a sufficient union region for decoding. Even when multiple adjacent instances are merged into a single connected region, the decoder still receives signals corresponding to the same watermark rather than conflicting messages, so correct decoding can still be achieved. By contrast, in the 1 instance case, the smaller watermark region is more sensitive to localization errors. These results indicate that, when all instances share the same watermark, the difference between with and without the instance segmentation branch is not significant.

However, in the practical instance-level multi-watermarking scenario, adjacent instances may carry different watermark messages. Without the segmentation branch, the localization result alone cannot separate such adjacent watermark regions, and the decoder would receive mixed signals from different instances, making independent decoding difficult. Therefore, the instance segmentation branch is essential for enabling true instance-level separation and independent decoding when different instances carry different watermarks.

H. Comparison of Different Segmentation Networks

The pretrained instance segmentation network is used to generate instance masks during inference and is not jointly trained with the watermark encoder or decoder. Thus, the framework is not tied to a specific segmentation architecture, although the quality of the generated masks may affect mask fusion and decoding performance. To examine the stability of the proposed framework, we replace SOLOv2 [33] with two representative instance segmentation networks, Mask R-CNN [31] and CondInst [34]. The former is a classical two-stage framework that first detects instances and then predicts their masks, while the latter is a one-stage method that generates instance masks using dynamic convolution. We report results under both full and partial screen-shooting with varying numbers of watermarked instances.

As shown in Table IX, the framework maintains consistently high extraction accuracy, remaining above 96%, when different segmentation networks are used. Under both full and partial screen-shooting, SOLOv2, Mask R-CNN, and CondInst all achieve strong performance, with only minor variations across different numbers of watermarked instances. The performance gap among the three models is limited. These results demonstrate that the proposed instance-level decoding framework remains stable across different segmentation architectures rather than relying on a specific model.

V. DISCUSSION

Although the proposed framework demonstrates strong robustness and flexibility, certain limitations should be noted. Watermark embedding is inherently restricted to instance regions. Consequently, no watermark can be recovered if the

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TABLE IX
EXTRACTION ACCURACY WITH DIFFERENT INSTANCE SEGMENTATION NETWORKS

Setting	Full Screen-shooting					Partial Screen-shooting				
	1 Instance	2 Instances	3 Instances	4 Instances	5 Instances	1 Instances	2 Instances	3 Instances	4 Instances	5 Instances
SOLOv2 [33]	99.67%	99.46%	99.09%	99.05%	98.73%	98.40%	98.68%	98.24%	96.15%	97.86%
Mask R-CNN [31]	98.55%	99.10%	98.59%	99.70%	98.01%	98.08%	98.27%	98.89%	96.71%	97.40%
CondInst [34]	98.32%	99.06%	98.67%	99.68%	98.79%	96.43%	98.55%	98.26%	98.50%	98.47%

captured area contains no object instances. However, this limitation has limited impact in practice, as the semantic and copyright value of images primarily resides in object instances, while background-only captures are usually less meaningful. In real-world deployments, natural images may not inherently possess instance masks. For large-scale applications, these masks can be obtained automatically using pretrained instance segmentation networks, while manual refinement with mainstream image editing tools such as Photoshop, Canva, or Pixso can be optionally used for images requiring precise control. The segmentation network introduces an additional inference cost, and its mask quality may affect mask fusion and decoding performance. It should also be noted that the framework is mainly applicable to common natural image scenarios with sufficiently recognizable instances and moderate capture quality. Its performance may degrade in more challenging cases, such as non-natural images, highly cluttered scenes with ambiguous instance boundaries, or situations where only partial instance fragments are captured. In particular, when the captured region contains only a small fragment of a watermarked instance, decoding may fail because the embedded watermark information is incomplete. Nevertheless, under most common practical scenarios, the proposed framework still maintains strong robustness and reliable decoding performance. At present, instance masks within the same image are constrained to be spatially disjoint to avoid signal interference. In scenes with overlapping or heavily occluded objects, overlapping pixels are assigned to only one instance during preprocessing. This assignment may make the affected occluded instance more sensitive to localization and segmentation errors if its remaining visible region becomes very small. Therefore, the current implementation is more suitable for natural images with distinguishable object instances, while scenes with severe object overlap may benefit from more advanced instance segmentation or conflict-aware embedding strategies.

In future work, we aim to improve the adaptability and interactivity of the framework. To this end, we will develop more flexible embedding strategies capable of handling partially overlapping instances, such as adaptive region weighting or conflict-aware watermark modulation. These enhancements will enable more adaptive watermark embedding in complex practical scenarios.

VI. CONCLUSION

In this paper, we propose an instance-level multi-watermarking framework that is robust against partial screen-shooting. Unlike existing image-level watermarking methods, our method enables fine-grained, region-specific watermark embedding at the instance level. To achieve controllable multi-watermark embedding, we designed a mask-gated adapter, which fuses instance masks with watermark information, enabling flexible control over the number and positions of embedded watermarks. During decoding, we introduce a parallel extraction architecture that integrates a watermark locator and an instance segmentation network. The two branches complement each other: the locator identifies the watermark regions, while the instance segmentation network distinguishes individual instances. Their fusion enables precise localization of watermarked instances and facilitates accurate watermark decoding. Extensive experiments demonstrate that our method achieves competitive performance in imperceptibility and robustness. In particular, it exhibits strong resistance to instance-level partial screen-shooting and supports flexible, fine-grained multi-watermark embedding.

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